PHYSICAL CUMULATIVE EXAM
MARCH 9, 2001

$$MnO_2\,(s) + FeCl_2\,(aq) + HCl\,(aq) \rightarrow MnCl_2\,(aq) + FeCl_3\,(aq) + H_2O\,(l)$$

(1) Express above redox reaction in two reduction half-reactions.

(2) Calculate the standard cell potential and the equilibrium constant for the reaction. The standard reduction potential of $MnO_2\,(s)/Mn^{2+}(aq)$ redox pair under acidic conditions is 1.23V. The standard reduction potential of $Fe^{3+}\,(aq)/Fe^{2+}\,(aq)$ redox pair is 0.77V.

(3) The Nernst equation is $E = E° - (RT/\nu F)\ln Q$. E and E° are the cell potential for given conditions and the standard conditions, respectively. R: the gas constant. T: Kelvin temperature. ν: number of electrons transferred between the reduced and oxidized species in a given reaction. F: the Faraday constant. Q: the concentration product of the reaction. Calculate F, the Faraday constant.

(4) Calculate the cell potential at pH = 0 and pH = 3, assuming all other substances are at their standard states.

Constants may be needed:

Avogadro constant = 6.0×10^{23} (mol^{-1})
Elementary charge (charge carried by a single electron) = 1.60×10^{-19} (C).
Gas constant = 8.31 ($J\,K^{-1}\,mol^{-1}$)
Speed of light = 3.00×10^{8} ($m\,s^{-1}$)
Planck constant = 6.63×10^{-34} (J s)